Measuring spiral structure and testing its incremental information for galaxy redshift estimation

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Abstract

Resolved spiral structure is an accessible source of information about galaxies, but its usefulness for locating galaxies must be measured separately from the ability to recognize their arms. We present a compact, interpretable pipeline that combines calibrated three-band images, multiscale spatial filters and human arm-region annotations, then tests whether the resulting structure summaries improve catalog-redshift prediction. Sky groups separate training, calibration and testing. On 29 test galaxies, the arm-region classifier reaches mean average precision 0.720, compared with 0.655 for a capacity-matched brightness/radius comparator; the equal-group gain is 0.0848, with a 95% bootstrap interval of 0.0475–0.1247. Applied without retraining to additional strong spirals in previously excluded sky regions, it reaches 0.689 on 57 evaluable galaxies. Eight predicted-region summaries, however, do not establish an incremental redshift gain over photometry, apparent geometry and simple image descriptors. The equal-group reduction in absolute redshift error is −0.000471 on the original 29 galaxies and +0.000440 on 59 additional galaxies; both intervals include zero. Stronger retrospective morphology baselines leave this conclusion unchanged. A separate human-centerline experiment shows why region agreement cannot be used as a substitute for validated pitch or width measurements. The pipeline supplies useful region measurements and a controlled test of their downstream information, while keeping angular structure, statistical redshift and physically calibrated distance distinct.

Keywords: galaxies: spiral — methods: image processing — methods: statistical — techniques: photometric — distance scale

1 Introduction

Spiral arms are among the most recognizable structures in astronomical images. Their winding, separation, contrast and colors can be measured without first assigning a mass model to the galaxy. A practical measurement system could therefore support studies of disk structure, provide regions for spatially resolved spectroscopy, and offer additional predictors for statistical redshift inference. These applications ask different questions. Recognizing where people draw an arm is a region-classification problem; determining its centerline or pitch is a geometric problem; estimating redshift is a supervised inference problem. Reliable progress requires each link to be evaluated on its own.

Automatic arm measurement already has a substantial literature. Davis and Hayes [2014] introduced SpArcFiRe, which extracts arm segments and fits logarithmic spirals. Hart et al. [2017] combined these measurements with Galaxy Zoo classifications to investigate arm formation. Citizen-science projects supply complementary information: Galaxy Zoo 2 describes detailed visual morphology [Willett et al., 2013]; Galaxy Zoo:3D supplies spatial masks of arms, bars and foreground stars [Masters et al., 2021]; and Galaxy Builder provides multicomponent models and human-guided arm traces [Lingard et al., 2020, 2021]. More recently, the ZooBot:3D preprint describes large-scale neural arm/bar segmentation in Legacy Survey images [Spindler et al., 2026]. These studies motivate a focused question about the *information contributed by an explicit arm representation*, rather than a claim to have invented arm detection.

Morphology can also help photometric-redshift estimation. The benefit depends on the photometric bands, sample and representation. Soo et al. [2018] studied size, ellipticity, Sérsic index and surface

brightness, finding greater improvements when fewer bands were available. Pasquet et al. [2019] demonstrated redshift prediction directly from multiband images. An arm-specific feature may be useful even when global morphology is already supplied, but this must be tested against a baseline with comparable model capacity and measurement quality. Otherwise, an apparent improvement could reflect brightness, apparent size, resolution or observational selection rather than spiral structure.

This paper makes three connected contributions. First, it presents a transparent three-band measurement pipeline with angular filter scales, explicit masks and separate artifacts for visualization and downstream inference. Second, it quantifies region-ranking performance against human annotations, including a spatially distinct additional cohort. Third, it measures the incremental value of eight predicted-region summaries for redshift, with common regressors, shared quality covariates and uncertainty calculated at the sky-group level. Separate geometric and dynamical analyses establish which further physical interpretations would require additional evidence. The software is intended to make these distinctions operational, so that a visually convincing map does not silently become a claimed physical measurement.

2 What an image can measure

2.1 Angular structure, physical size and redshift

For a small angular separation θ, a transverse physical separation is

$$L = D_A\theta, \qquad \theta \text{ in radians}, \tag{1}$$

where D_A is angular-diameter distance [Hogg, 1999]. A digital separation s pixels becomes $\theta = sp$ for a local angular scale p. This determines an angle, not a distance: the same angular pattern can arise from different intrinsic sizes at different distances. Similarly, an apparent flux constrains luminosity only through a distance-dependent conversion. In the nearby Euclidean limit, simultaneous changes $D \mapsto \lambda D$, $L_{\text{size}} \mapsto \lambda L_{\text{size}}$ and $L_{\text{luminosity}} \mapsto \lambda^2 L_{\text{luminosity}}$ preserve apparent size and total flux. Population relations, spectral information and selection can make some such combinations more probable than others. A learned redshift estimator uses those calibrated statistical relations.

Redshift and distance are consequently separate outputs. The target in this study is a catalog spectroscopic redshift, z. A cosmological conversion would require a cosmological model, and local peculiar velocities also affect the relation between observed redshift and expansion distance. We report errors in z and do not reinterpret them as independently validated distance errors. Physical arm separations can be reported when an external distance is supplied, with that distance uncertainty propagated.

2.2 Projected and deprojected spiral geometry

Let (x, y) be angular coordinates relative to a supplied center. Rotating by the disk position angle gives (x', y'). For a thin circular disk inclined by i,

$$X = x', \qquad Y = \frac{y'}{\cos i}, \qquad R_\theta = (X^2 + Y^2)^{1/2}, \qquad \phi = \operatorname{atan2}(Y, X). \tag{2}$$

The approximation becomes sensitive to geometry errors at high inclination. It is appropriate only after specifying what defines the disk plane; a bar, bulge or foreground object can bias a moment-based estimate.

A logarithmic arm segment obeys

$$R_\theta(\phi) = R_0 \exp[b(\phi - \phi_0)], \qquad |\psi| = \arctan|b|, \tag{3}$$

where ψ is the pitch angle relative to the local circular direction. Pitch is dimensionless and does not require a distance, but its measured value depends on the center, inclination, position angle and

selected arm segment. Different arms in one galaxy need not share one pitch [Davis and Hayes, 2014, Lingard et al., 2021]. A useful region map alone does not determine these geometric quantities. For this reason the primary classifier works in a fixed *projected circular annulus*; deprojection is reserved for a separate centerline validation with externally supplied disk geometry.

2.3 Passbands and spatial filters

A passband measures a different part of the spectral energy distribution. A spatial filter transforms measurements already present in an image. We use the calibrated g, r, z science planes individually, preserving their different brightness and color information. A single-band image is naturally a grayscale measurement. Converting a rendered color image into black and white changes its display representation; it does not supply a new passband or restore calibrated fluxes that were lost during rendering.

For a band image I_b, write its radial-profile residual as $J_b = I_b - B_b - m_b(R_\theta)$, where B_b is a robust background and m_b a radial median profile. Gaussian scale space is

$$L_b(\boldsymbol{x}; \sigma) = G_\sigma * J_b, \qquad G_\sigma(\boldsymbol{x}) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{|\boldsymbol{x}|^2}{2\sigma^2}\right]. \tag{4}$$

Differences of smoothed images emphasize local contrast; gradients locate changes; Hessian eigenvalues distinguish elongated bright ridges from locally isotropic structure. Scale-normalized derivatives follow standard scale-space reasoning [Lindeberg, 1998], and the ridge descriptor is related to established Hessian-based enhancement [Frangi et al., 1998]. Our scales are fixed angular standard deviations of 0.4, 0.8 and 1.6 arcsec. They are measurement choices, not inferred physical arm widths. The full feature definitions and treatment of noise, boundaries and masks appear in the Supplementary Material.

3 Observations and experimental design

The original sample contains 160 resolved strong spirals selected deterministically from the public Galaxy Zoo:3D catalog, excluding 32 earlier feasibility objects. The spiral-vote fraction is at least 0.7 and the source selection requests one center. Native Legacy Survey DR9 g, r, z images, inverse-variance planes and local point-spread functions (PSFs) provide the science measurements [Dey et al., 2019]. Human spatial labels and centers come from Galaxy Zoo:3D [Masters et al., 2021]. Observed catalog *ugriz* photometry, apparent geometry and the heliocentric `NSA_Z` target are matched by identity to MaNGA DRPall, within the SDSS data framework [York et al., 2000, Bundy et al., 2015, Abdurro'uf et al., 2022].

Ten-degree right-ascension/declination tiles define sky groups. There are 91 training galaxies in 38 groups, 40 calibration galaxies in 13 groups and 29 test galaxies in 13 groups. No group crosses these boundaries. Spatial grouping addresses a common weakness of random splitting in structured data [Roberts et al., 2017]; it is not a proof that all remaining groups have independent calibration errors. Training is used for fitted coefficients and model selection, calibration for a visualization sigmoid and prediction intervals, and testing for the final original comparison.

An additional acquisition excludes all 192 earlier identifiers and all 67 earlier sky tiles. It contains 99 objects in 39 new tiles: 59 eligible strong spirals in 30 groups, plus 40 weak-spiral or low-vote comparison objects. The 59 strong spirals form the principal additional redshift cohort. All are evaluated with the original frozen models, without threshold or calibration updates. This tests new sky regions within the same imaging survey.

Coverage is retained explicitly. One additional strong spiral has no usable native science/PSF product after repeated requests. Another lacks both scoring classes after masking. Consequently arm AP is defined for 57 of 59 strong spirals; catalog-redshift predictions retain all 59 rows using

training-fitted imputation when needed. A low-vote or unasked arm task is treated as unknown, not as a verified arm-free control.

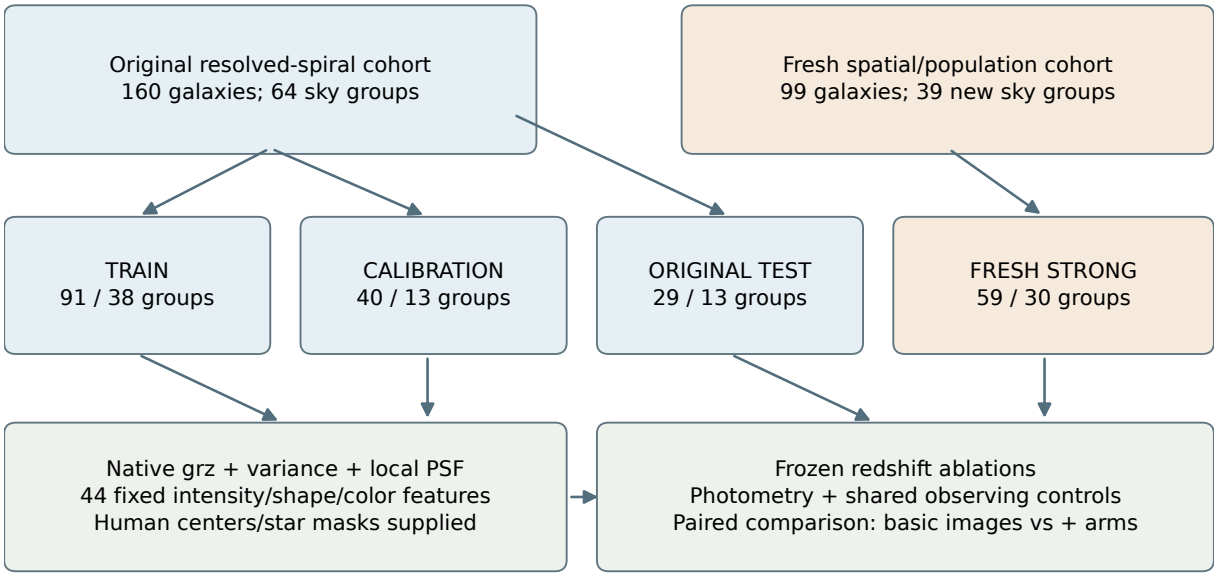


Figure 1: Measurement and evaluation design. The image-to-region model and the redshift regressors are trained using only training sky groups. Display calibration is separate from the raw-score representation used for redshift. The additional cohort is evaluated with the frozen transformations and models.

4 Measurement and inference methods

4.1 Registered science measurements and reference regions

Each galaxy is measured within a fixed 3–23 arcsec annulus about its supplied human center. Foreground-star labels exclude contaminated pixels. All bands use the same native grid: only the reference labels are mapped through WCS by nearest neighbor. The reader checks the target-containing image brick, band, image/inverse-variance pairing and coordinate registration. PSF matching broadens each band to the worst local Gaussian moment-equivalent width. A shared guard excludes pixels within four times the largest filter scale of masked holes or boundaries, in addition to the convolution mask treatment.

The reference-positive class is a pixel covered by at least three human arm polygons. Other scored pixels form the comparison class. This is an annotation agreement target: an unmarked pixel can contain an incompletely traced arm. Five-vote labels provide a sensitivity analysis, with the changing positive prevalence and evaluable sample reported separately.

4.2 A compact multiscale region classifier

The 44 local features comprise three radius terms, 13 intensity/contrast/shape terms in each of three bands and two signed color terms. They include radial profile and residual brightness, differences of Gaussians, gradients, a Laplacian-like response, bright-ridge strength and ridge orientation relative to the center. None requires a redshift, physical size or mass model.

Seven fixed logistic/ExtraTrees candidates are compared by three-fold sky-group cross-validation within the training set. ExtraTrees uses randomized tree ensembles [Geurts et al., 2006], implemented with scikit-learn [Pedregosa et al., 2011]. The selected full model has 80 trees, maximum depth 12, minimum leaf size 32 and a feature fraction of 0.7. Each training galaxy contributes at most 400 positive and 400 comparison pixels, with equal total weight per galaxy. Thus a galaxy with more image pixels cannot dominate simply because it covers more area.

The brightness/radius comparator has the same eligible logistic/tree capacities but only three radius terms, three normalized band intensities and the g-band radial profile. Training cross-validation also selects its depth-12 tree model. Raw g intensity and a circular-profile score provide simpler comparisons. This design asks whether local multiscale features add beyond intensity and radius under a controlled learner budget.

The visualization artifact applies a positive-slope sigmoid to the logit of the tree score, following the calibration approach associated with Platt [2000]. Calibration data select one display threshold, 0.30. Average precision (AP) evaluates score ranking and is unaffected by the monotone calibration. We compute AP separately for each galaxy and average it, avoiding a pooled-pixel metric dominated by large objects. Fixed-threshold precision, recall, F1 and intersection-over-union are secondary region diagnostics.

4.3 A separate representation for downstream redshift

For downstream inference we save the training-only classifier *without* the display sigmoid, using a fixed raw threshold of 0.5. Training galaxies receive descriptors from a classifier fitted without their entire sky group; calibration and test galaxies receive descriptors from the complete training-only model. The summary annulus depends on the image masks, not on human arm coverage. Calibration arm labels therefore do not enter redshift features.

Let $s_j \in [0, 1]$ be this raw score at valid pixel j and u_j its radius divided by 23 arcsec. The eight arm summaries are the mean, 90th percentile and standard deviation of s, the score-weighted mean of u, mean scores inside and outside $u = 0.55$, mean binary entropy and the fraction with $s \geq 0.5$. The raw score ranks agreement with the annotation protocol; balancing classes during training means it is not a calibrated physical arm probability.

Four feature families share one regression design. P contains five observed magnitudes and four adjacent colors. PG adds angular half-light radius and axis ratio. PGI adds seven simple image descriptors: three annular positive fluxes, a radial half-flux scale, residual contrast and two colors. PGIA adds the eight arm summaries. All share magnitude-error, PSF-width and image-noise covariates, with consistent missing-feature handling. This reduces the opportunity for arm availability or resolution to act as an exclusive proxy for redshift.

Each regressor uses 128 ExtraTrees, minimum leaf size four and the same seed, with training-median imputation and missingness indicators. The fitted target is $t = \log(1 + z)$ and predictions are transformed back through $\hat{z} = \exp(\hat{t}) - 1$. Training cross-validation selects P as the retained deployment family. The scientific arm contrast remains PGI versus PGIA, regardless of this choice. No original or additional test result changes the deployment selection.

4.4 Paired effects and uncertainty

For a sky group g containing n_g galaxies, define

$$d_g = \frac{1}{n_g} \sum_{i \in g} \left[|z_i - \hat{z}_{i,\mathrm{PGI}}| - |z_i - \hat{z}_{i,\mathrm{PGIA}}| \right], \qquad \Delta = \frac{1}{G} \sum_{g=1}^{G} d_g. \tag{5}$$

Positive Δ favors the arm summaries. Equal-group weighting differs deliberately from galaxy-weighted mean absolute error (MAE). Percentile bootstrap intervals resample whole groups [Efron, 1979]; the

uncertainty unit is not a pixel. The two cohorts are reported separately. The final statistical analysis uses 100,000 group resamples and also examines a Student-t interval as a small-sample sensitivity.

For interpretable effect sizes we additionally examine a reduction $\delta_z = 0.001$. This is a retrospective practical margin for this pilot, not a physically validated universal threshold. A confidence interval that crosses zero is inconclusive about the sign. To support practical equivalence at this margin, the relevant interval would need to be contained within $[-\delta_z, \delta_z]$; a nonsignificant difference alone is insufficient.

Nominal 90% split-conformal intervals use the maximum absolute residual in t within each calibration sky group [Lei et al., 2018, Angelopoulos and Bates, 2021]. With 13 groups the finite-sample rank is the largest calibration group residual. These intervals concern catalog redshift in comparable exchangeable groups. Distribution shift requires separate treatment [Tibshirani et al., 2019].

5 Results

5.1 Multiscale features improve agreement with arm regions

The original test classifier reaches galaxy-macro AP 0.7197, versus 0.6551 for the capacity-matched brightness/radius comparator, 0.6267 for raw g brightness and 0.4904 for the circular-profile score. The improvement is not limited to a visibly selected example: the equal-sky-group AP gain over the matched comparator is 0.0848, with a 95% bootstrap interval of 0.0475–0.1247. At the fixed display threshold, macro precision is 0.590, recall 0.678 and F1 0.581. These results establish a useful ranking improvement for the specified region task and selected population.

Without retraining, mean AP is 0.6890 on the 57 evaluable additional strong spirals, with a sky-group bootstrap interval of 0.6409–0.7376. Median fixed-threshold F1 is 0.579. This provides evidence that the region-ranking performance extends to additional sky regions in the same survey. It does not validate every output as a complete physical arm.

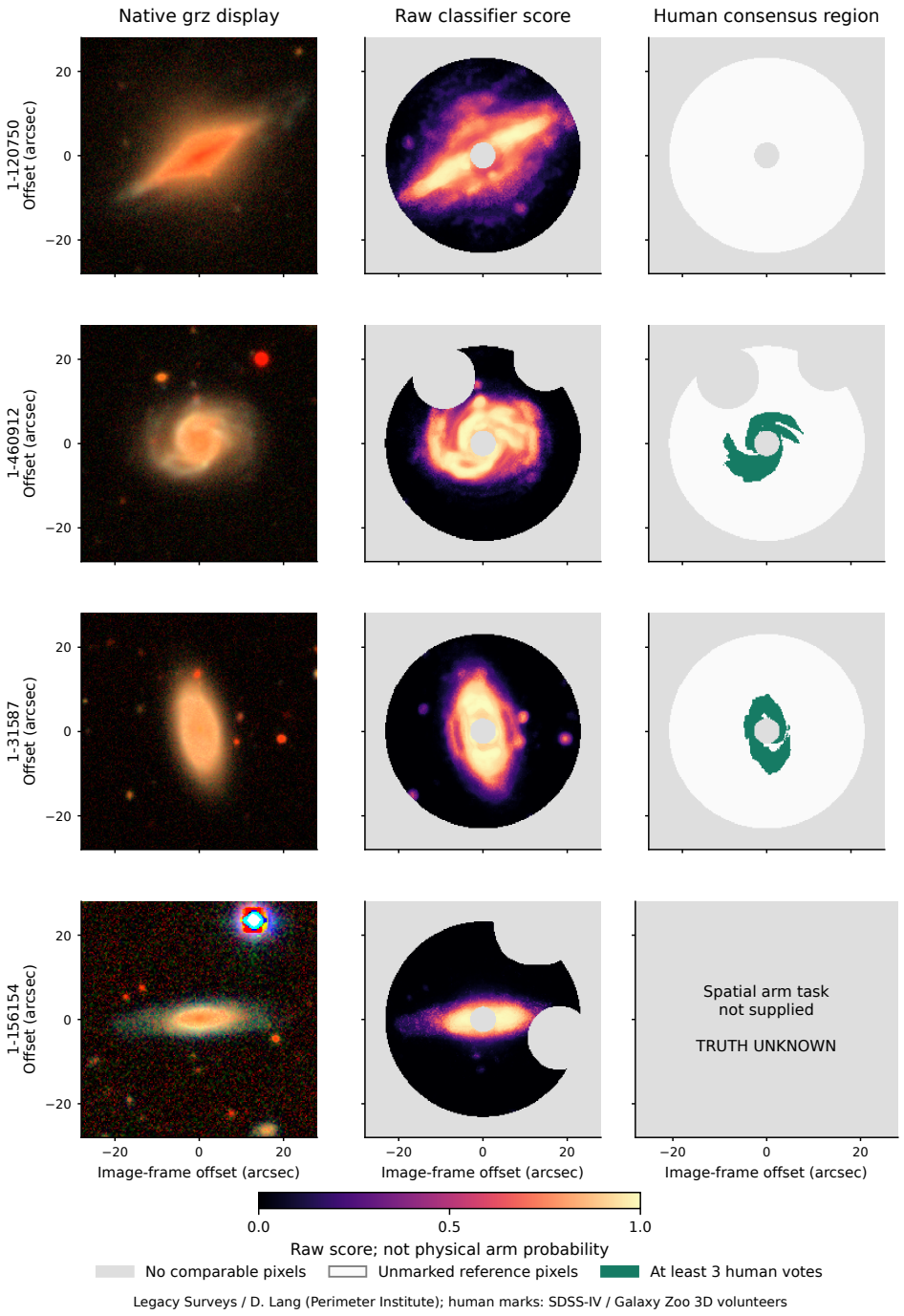


Figure 2: Fixed science-image, raw-score and reference panels: the first two additional strong spirals, first weak/bar-proxy object and first unasked control in the pre-scoring gallery. Scores use the training-only uncalibrated branch; they are distinct from geometric centerlines or physical arm probabilities. The no-consensus case is retained, and an unasked task has unknown truth. White valid unmarked pixels are comparison labels, not verified physical absence. Legacy Surveys / D. Lang (Perimeter Institute). Human region references: SDSS-IV and Galaxy Zoo:3D volunteers [Masters et al., 2021].

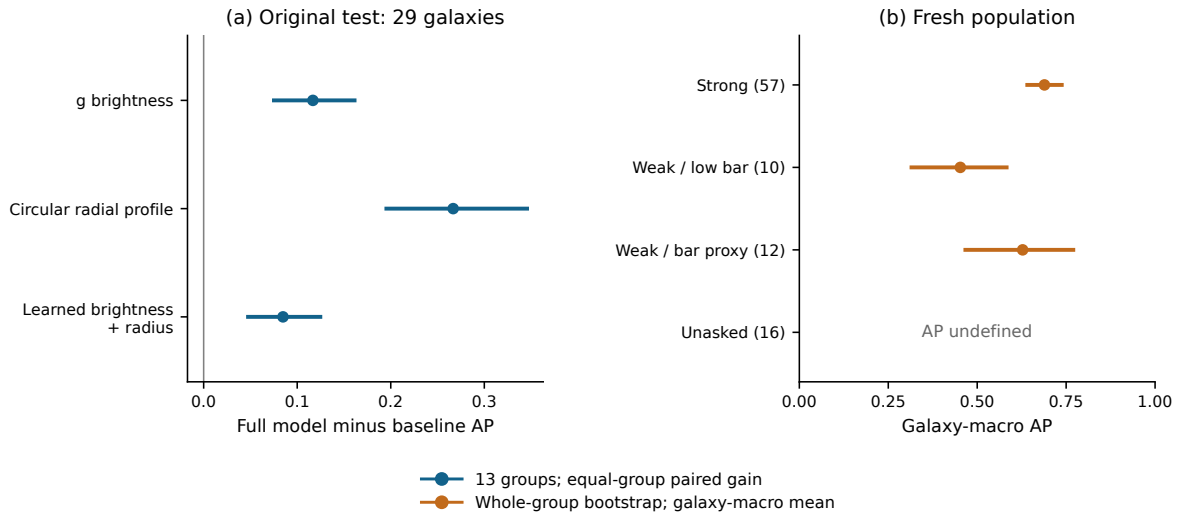


Figure 3: Arm-region validation. Original-cohort comparisons use the same scoring pixels and independently selected matched-capacity baseline; additional cohort performance uses the frozen full classifier. Galaxy-macro AP and equal-sky-group contrasts are distinct estimands. Reference-positive pixels require at least three human polygons.

5.2 The incremental redshift effect remains unresolved

Table 1 reports galaxy-weighted MAE for all four frozen families. The original-cohort PGI-to-PGIA contrast is negative at equal-group weighting, $\Delta = -0.000471$, with a 95% interval $[-0.001391, 0.000367]$. On the additional 59 strong spirals it is positive, $\Delta = 0.000440$, with interval $[-0.000555, 0.001452]$. Both include zero. The original interval excludes a gain as large as the retrospective margin of 0.001, while the additional-cohort interval still permits a gain at that margin. Neither establishes practical equivalence.

The P family, retained by training selection, has original coverage 29/29 with median redshift-interval width 0.08954. On the additional cohort, coverage is 57/59 and median width 0.09043. The corresponding whole-group coverage is 28/30. These broad intervals are informative about limited precision, even when empirical coverage is high. Input-only support checks flag 15/59 additional P predictions for possible abstention. Table 1 keeps all rows instead of reporting only favorable accepted predictions.

Table 1: Catalog-redshift prediction by the original frozen regressors. MAE weights galaxies equally; the principal arm contrast in equation (5) weights sky groups equally. All 59 additional rows remain in the analysis, including the missing-image row.

Feature family	Original: 29 galaxies	Additional: 59 galaxies
P: photometry	0.009635	0.013190
PG: P + apparent geometry	0.009469	0.012560
PGI: PG + simple image summaries	0.009266	0.012767
PGIA: PGI + arm-region summaries	0.009751	0.012532

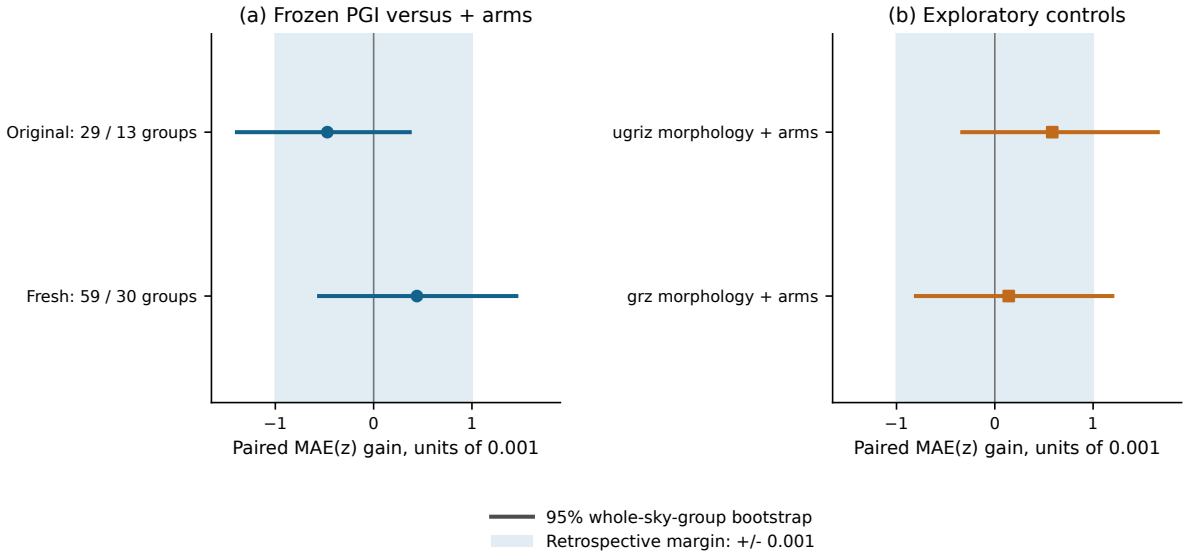


Figure 4: Paired reductions in absolute catalog-redshift error from adding arm-region summaries. Positive values favor arms. The original and additional cohorts use frozen original models. L5/L3 comparisons are retrospective controls. Intervals resample whole sky groups. The shaded practical margin ±0.001 was introduced retrospectively and does not convert this pilot into a confirmatory equivalence study.

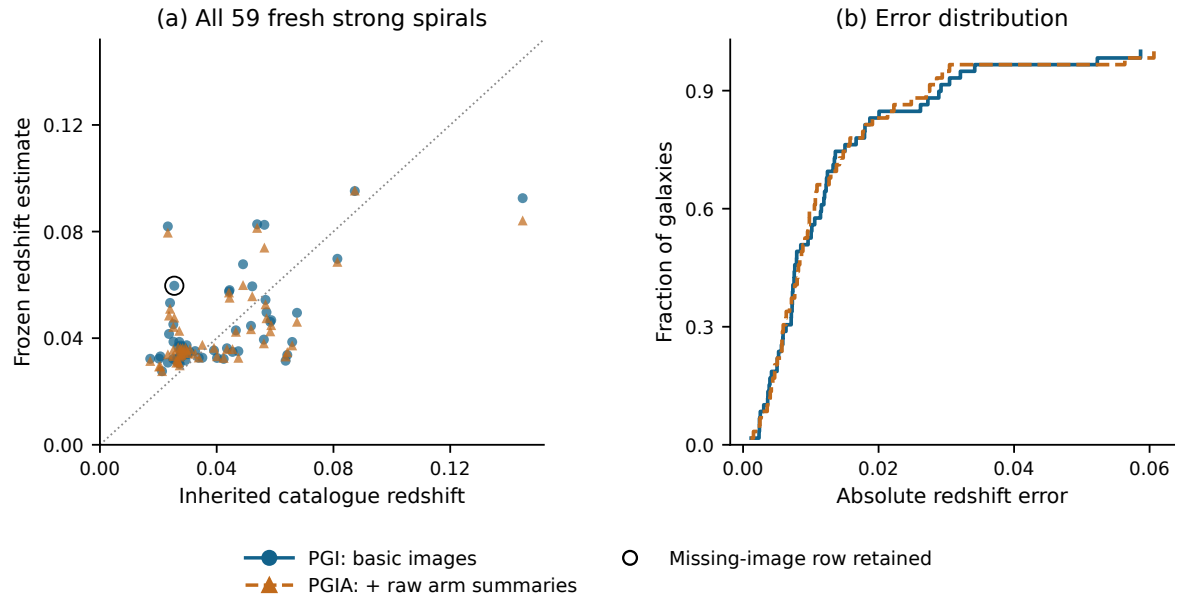


Figure 5: Frozen PGI and PGIA predictions and absolute-error cumulative distributions for all 59 additional strong spirals. The missing-image row is imputed and marked, and no input-support screening removes rows. The galaxy-weighted MAEs differ from the equal-group contrast in Figure 4. These are catalog redshifts, not independent geometric distances.

5.3 Stronger photometric morphology controls

We performed an additional, explicitly retrospective baseline analysis motivated by Soo et al. [2018]. L5 contains observed five-band photometry, apparent geometry, observed r-band surface brightness and Sérsic index, plus the shared quality block. L5A adds the same eight arm summaries. L3/L3A restrict explicit photometry and magnitude errors to g, r, z, retaining the specified angular structure and grz image-quality information. All use the same 91 training galaxies and common ExtraTrees capacity; the additional cohort does not select new hyperparameters.

On 59 additional strong spirals, equal-group MAE falls from 0.013132 to 0.012547 for L5/L5A, a gain of 0.000584 with interval $[-0.000331, 0.001658]$. For L3/L3A it falls from 0.012928 to 0.012786, a gain of 0.000142 with interval $[-0.000802, 0.001195]$. These controls leave the incremental arm result unresolved. They are a controlled implementation of literature-motivated observable features, not a reproduction of the Morpho-z algorithm, training population or survey. The three-band restriction also does not constitute a cross-survey experiment.

5.4 Centerlines and pitch require their own validation

A separate 24-object Galaxy Builder sample supplies genuine human centerlines and fitted disk geometry [Lingard et al., 2020, 2021]. Twelve development objects select one classical logarithmic-template configuration; twelve test objects assess localization within the supplied PSF tolerance. Test mean centerline F1 is 0.223, with a whole-galaxy interval of 0.105–0.353. The extractor predicts 70 segments against 19 retained reference segments; seven spatially matched segments have median absolute pitch error 11.42 degrees. This fails the development goal of F1 at least 0.7 and matched pitch error at most 5 degrees. It is a different experiment from the supervised region classifier and should not be interpreted as its centerline accuracy.

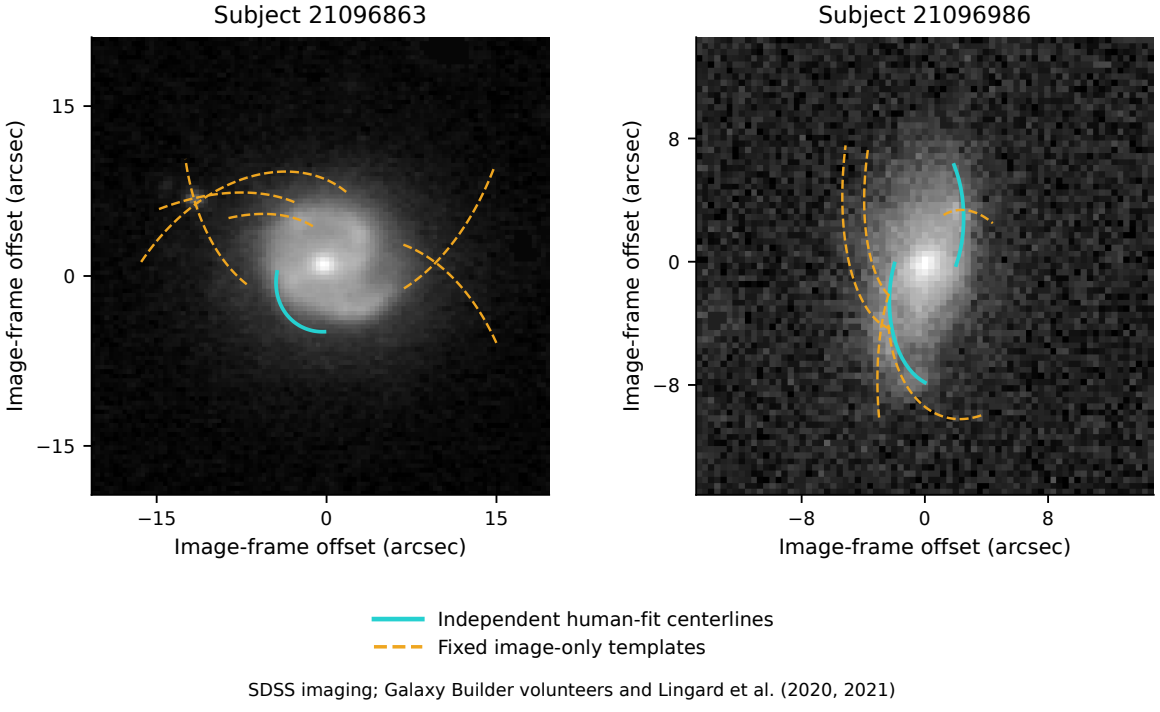


Figure 6: Separate human-centerline validation of the classical geometry prototype. Region labels and centerlines answer different questions. Pitch error is conditional on the seven matched segments, while localization F1 includes unmatched predictions. SDSS imaging; Galaxy Builder volunteers and author-fitted references [Lingard et al., 2020, 2021].

An archival comparison uses actual published SpArcFiRe-related galaxy pitches from the Hart et al. [2017] catalog. All 24 Galaxy Builder identifiers match; 13 have finite reliable-arc pitches and

11 retain the published missing sentinel. On the six measured test identities, the mean absolute difference from the Galaxy Builder reference summary is 7.73 degrees, with a conditional interval of 5.06–10.69 degrees. The two galaxy summaries weight different selected arcs, and this is a catalog comparison rather than a rerun of the author pipeline. It supports attention to reference conventions and missing coverage, not a ranking of the present extractor against SpArcFiRe.

6 Interpretation and applications

The clearest positive result is that a compact multiscale representation improves region ranking beyond brightness and radius under a matched learner budget. Its constituent measurements have a direct use: arm-associated pixels can be compared with interarm regions, cross-matched to resolved spectroscopy, or used to prioritize objects for geometric modeling. The frozen additional cohort demonstrates useful same-survey transfer of that region task.

The redshift experiment addresses a further question, and its answer is more limited. Once colors, apparent geometry and quality information are present, the eight regional summaries have no established additional value in these cohorts. This does not negate the region measurement. Nor does it bound all information in an image: the summaries may discard topology, continuous centerlines, segment-to-segment pitch variation or stellar-population structure. The result concerns a defined representation and regression family.

The physical distinction is equally useful. A rotation curve constrains the gravitational field through $v_c^2 = R\,\partial_R\Phi$; an image measures projected light. Combining them requires an explicit light-to-mass calibration, disk geometry and independently measured velocities. Established Tully–Fisher methods illustrate this additional information [Tully and Fisher, 1977, Schombert et al., 2020]. In the Supplementary Material we provide a finite pressure-supported equilibrium calculation, derived from known lowered-isothermal models [King, 1966, Gieles and Zocchi, 2015], to keep dynamical interpretations physically consistent. This is a separate conditional model, not a source of missing distance labels or a viscosity measurement from pixels.

7 Limitations and a focused route forward

The samples contain resolved, strongly selected nearby spirals. Human centers, foreground masks and classifications remain external inputs. The additional cohort changes sky regions within the same survey; it does not establish transfer to another instrument, arbitrary morphology or unresolved galaxies. Incomplete human regions define a reproducible annotation task, but not complete physical truth or a verified real arm-free negative population.

Gaussian moment matching approximates the PSF and does not fully account for non-Gaussian wings or noise covariance. Fixed angular filters probe different physical scales across the sample. The finite annulus omits central and outer structure, while star/border guards can remove substantial areas. These choices are measured coverage conditions rather than proof of complete arm recovery. Observed width accuracy has not been validated independently.

Statistical resolution is set by the 13 and 30 test groups, not the number of pixels. Conditional bootstrap intervals omit retraining uncertainty, annotation systematics and uncertainty in the selected population. The cached out-of-fold representation protects training descriptors from their own group, but the training-only choice of upstream representation is not fully nested within every downstream cross-validation fold. Practical margins, the stronger baselines and the archival comparison were introduced after initial results; they should guide a new study rather than be described as prospective tests.

A next confirmatory study should fix its feature representation, survey population, practical margin and missing-data policy before examining new redshifts. It should include a genuine same-input comparison to mature arm methods, with training-overlap checks for pretrained models such as

ZooBot:3D [Spindler et al., 2026]. A useful geometric extension must first pass independent centerline and width tests. Conditional planning from the observed group dispersion suggests roughly 66 independent groups for 80% power to detect a true 0.001 mean gain against zero, rising to 106 under an upper-variance scenario. These calculations assume stable frozen models and comparable groups; they are not evidence that such a gain exists or a promise of sample sufficiency.

8 Conclusions

The three-band multiscale classifier provides a measurable improvement in human arm-region agreement over a matched brightness/radius baseline, and retains useful performance on additional sky regions. This establishes a practical region-analysis component for resolved spirals. Its eight regional summaries do not yet establish an improvement in catalog-redshift prediction; the original and additional grouped contrasts remain statistically inconclusive, including after stronger observable morphology controls. Separate centerline results show that geometric accuracy requires additional development and independent validation. The resulting framework supports careful measurement and calibrated inference while preserving the distinction between apparent structure and physical distance.

Data and software availability

The companion source distribution at https://github.com/nalin-dhiman/Galaxy-arm-project contains the publicly available measurement code, frozen-model metadata, numerical result tables, figure-generation provenance and editable LaTeX sources. Original survey products are obtained from their credited public archives; access and reuse follow the relevant archive terms. The Supplementary Material describes the output schema and reproducibility sequence.

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